

Linear Algebra Multiplications: A Complete Guide

1 Matrix–Vector Multiplication (Ax)

Definition

If $A \in \mathbb{R}^{m \times n}$ and $x \in \mathbb{R}^n$, then $Ax \in \mathbb{R}^m$:

$$(Ax)_i = \sum_{j=1}^n A_{ij}x_j.$$

Mental Models

- 1) **Row perspective (dot products)** Each row of A is dotted with x . Example:

$$A = \begin{bmatrix} 2 & 1 \\ -1 & 3 \end{bmatrix}, \quad x = \begin{bmatrix} 4 \\ 5 \end{bmatrix}, \quad Ax = \begin{bmatrix} 13 \\ 11 \end{bmatrix}.$$

- 2) **Column perspective (linear combinations)** Each column of A is scaled by x_j , then summed:

$$Ax = 4 \begin{bmatrix} 2 \\ -1 \end{bmatrix} + 5 \begin{bmatrix} 1 \\ 3 \end{bmatrix} = \begin{bmatrix} 13 \\ 11 \end{bmatrix}.$$

- 3) **Transformation perspective** A acts as a linear transformation (rotation, scaling, shear). Example: $[1, 0]^T \mapsto [2, -1]^T$, $[0, 1]^T \mapsto [1, 3]^T$.

Questions to Ask

- Am I analyzing constraints/equations? (rows)
- Am I analyzing span/geometry? (columns)
- Am I thinking of a transformation? (basis action)

2 Vector–Matrix Multiplication ($x^T A$)

Definition

If $x^T \in \mathbb{R}^{1 \times m}$ and $A \in \mathbb{R}^{m \times n}$, then $x^T A \in \mathbb{R}^{1 \times n}$:

$$(x^T A)_j = \sum_{i=1}^m x_i A_{ij}.$$

Mental Model

x acts on rows of A , producing a weighted sum of rows.

Example:

$$x^T = [1 \ 2], \quad A = \begin{bmatrix} 3 & 4 \\ 5 & 6 \end{bmatrix}, \quad x^T A = [13, 16].$$

Questions to Ask

- Am I collapsing rows or columns?
- Is the result a row or column vector?

3 Vector–Vector Multiplication

Dot Product ($x^T y$)

$$x^T y = \sum_i x_i y_i \quad (\text{scalar}).$$

Example: $[1, 2, 3] \cdot [4, 5, 6] = 32$.

Outer Product (xy^T)

$$(xy^T)_{ij} = x_i y_j \quad (\text{matrix}).$$

Example:

$$\begin{bmatrix} 1 \\ 2 \end{bmatrix} \begin{bmatrix} 3 & 4 \end{bmatrix} = \begin{bmatrix} 3 & 4 \\ 6 & 8 \end{bmatrix}.$$

Questions to Ask

- Do I need a scalar similarity measure? (dot)
- Do I need a matrix of pairwise influences? (outer)

4 Matrix–Matrix Multiplication (AB)

Definition

If $A \in \mathbb{R}^{m \times n}$, $B \in \mathbb{R}^{n \times p}$, then:

$$(AB)_{ij} = \sum_{k=1}^n A_{ik} B_{kj}.$$

Mental Models

- 1) **Row–Column view:** each entry = dot product of row of A with column of B .
- 2) **Column view:** $AB = [Ab_1, Ab_2, \dots]$, i.e. A transforms columns of B .
- 3) **Composition:** AB = apply B first, then A .

5 Quick Shape Rules

$$\begin{aligned} Ax &: (m \times n)(n \times 1) \rightarrow (m \times 1), \\ x^T A &: (1 \times m)(m \times n) \rightarrow (1 \times n), \\ x^T y &: \text{scalar}, \\ xy^T &: (m \times 1)(1 \times n) \rightarrow (m \times n), \\ AB &: (m \times n)(n \times p) \rightarrow (m \times p). \end{aligned}$$

6 How to Read an Expression

- 1) Identify shapes (row/col).
- 2) Choose a mental model (rows, columns, transformation, composition).
- 3) Predict output (scalar, vector, matrix).

7 Practice Prompts

- (1) If A is 3×2 and x is 2×1 , what is $\text{shape}(Ax)$?
- (2) Write Ax as row dot products and as column combinations.
- (3) Compute $x^T A$ with $x = [1, 2]^T$, $A = \begin{bmatrix} 2 & 3 \\ 4 & 5 \end{bmatrix}$.
- (4) Show how the outer product can build a projection matrix.
- (5) If AB is defined, is BA always defined?

Final Takeaway

- Rows = equations/constraints.
- Columns = span/linear combinations.
- Transformations = geometry.
- Shapes = sanity check.